

CHE-MING, CHANG

cheming.chang1@gmail.com ◇ GitHub ◇ LinkedIn ◇ Los Angeles, CA

EDUCATION

Ph.D. student in Electrical and Computer Engineering, University of Southern California

Overall GPA: **4.00 / 4.00**

08/2025 - Expected 05/2030

Research Topic: AI/ML for EDA and heterogeneous integration

B.S. in Electrical Engineering, National Taiwan University

Overall GPA: **4.17 / 4.30**

09/2020 - 06/2024

Dean's List Award 2023 Spring

TSMC Semiconductor Advanced IC Design Program

Highlighted Coursework: Logic Synthesis and Verification, Physical Design for Nanometer ICs, Integrated Circuit Design, Computer-aided VLSI System Design, Digital Signal Processing in VLSI, Machine Learning

SKILLS

Programming C/C++, Python, Verilog, Tcl, Bash

EDA Tools Synopsys VCS / Design Compiler / PrimeTime / Library Compiler, Cadence Innovus

Languages Mandarin (Native), English (Proficient)

PUBLICATION

- [1] **Che-Ming Chang**, Shao-Yu Lo, Yao-Wen Chang and Sung Kyu Lim, "Encoding Symmetries into Operator Learning: Towards Accurate Multi-Physics Simulation in Advanced Packaging," in *Proc. of ASPDAC*, Jan 2027.
- [2] Pin-Han Chen, Chanhee Yang, Sungchul Joo, Hyungsoo Moon, **Che-Ming Chang**, Taek-Soo Kim and Sung Kyu Lim, "WAFER: Warpage Analysis via Fourier-Enhanced Residual Calibration for Large-Scale 2.5D IC Designs," in *Proc. of ICCAD*, November 2026, pp. 1-9.
- [3] **Che-Ming Chang**, Prashanth Vijayaraghavan, Charles Mackin, Ashutosh Jadhav, Hsinyu Tsai, Vandana Mukherjee and Ehsan Degan, "CODMAS: A Dialectic Multi-Agent Collaborative Framework for Structured RTL Optimization," in *Proc. of EACL Industrial Track*, March 2026.
- [4] Shao-Yu Lo, **Che-Ming Chang** and Yao-Wen Chang, "Advanced Packaging Warpage Modeling with DeepONet-Based Operator Learning," in *Proc. of ICCAD*, October 2025.
- [5] **Che-Ming Chang**, Jie-Hong Roland Jiang, Dah-Wei Chiou, Ting Hsu and Guin-Dar Lin, "Quantum Circuit Compilation for Trapped-Ion Processors with the Drive-Through Architecture," in *IEEE Trans. Quantum Eng.*, vol. 6, pp. 1-14, 2025.
- [6] Cheng-Yu Chiang, Yi-Hsien Chiang, Chao-Chi Lan, Yang Hsu, **Che-Ming Chang**, Shao-Chi Huang, Sheng-Hua Wang, Yao-Wen Chang and Hung-Ming Chen, "Mixed-Size Placement Prototyping Based on Reinforcement Learning with Semi-Concurrent Optimization," in *Proc. of ASPDAC*, January 2025, pp. 893-899.

RESEARCH EXPERIENCE

Southern California Computer-Aided Design (SCCAD) Lab

08/2025 - Present

Graduate Research Assistant, Advised by Prof. Sung Kyu Lim

Los Angeles, CA

- Researched in ML-based global routing optimization tasks.

Qualcomm, Methodology team

05/2026 - 08/2026

Research Intern, Advised by Mingyu Woo

San Diego, CA

- Enhanced internal ECO-stage Vt assignment algorithm integrated with Synopsys PrimeTime.
- Delivered up to 26% flow runtime improvement, 36% TNS improvement during leakage power optimization on SoC-scale production designs.

Electronic Design Automation Lab*Research Assistant, Advised by Prof. Yao-Wen Chang*

08/2023 - 01/2025

Taipei, Taiwan

- Assisted in RL for concurrent macro placement project, demonstrated 12.3% / 9% additional HPWL reduction on ISPD'05 benchmarks compared to DREAMPlace / modern ML-based placer GraphPlanner.
- Developed an operator learning framework to predict warpage in advanced packaging designs, achieving 435× speedup over 8-core finite element method while maintaining an average error of 1.9%.

IBM Almaden Research Center, AI for EDA group*Research Intern, Advised by Prashanth Vijayaraghavan*

06/2024 - 09/2024

San Jose, CA

- Developed a multi-agent framework for HDL code optimization by retrieving domain-specific knowledge into the code generation process, aided with compiler and simulator feedback.
- Achieved 20% / 10% improvements for larger LLMs /small fine-tuned LLMs compared to zero-shot and ReAct / Chain-of-Thought baseline on HDL optimization tasks.

Applied Logic and Computation in System Design Lab*Research Assistant, Advised by Prof. Jie-Hong Roland Jiang*

09/2022 - 06/2024

Taipei, Taiwan

- Proposed an analytical method to compile a QASM circuit onto a “drive-through” modular architecture by gate partitioning followed by a fidelity-aware placement.
- Achieved 10.8% / 9.6% higher fidelity compared to modern qubit mappers SABRE / t|ket>.

PROJECT EXPERIENCE

Approximate Logic Synthesis with Multi-Objective Simulated Annealing*Final Project in Logic Synthesis and Verification, GitHub*

10/2023 - 12/2023

Taipei, Taiwan

- Utilized multi-objective simulated annealing to synthesize approximate circuits within the given error bound based on the synthesis tool ABC.
- Demonstrated substantial area reduction (8-98%) over IWLS benchmarks within 1% error rate and shows quality (5-91%) and runtime improvements over the existing ALS tool ALSRAC.

QR decomposition engine for 5G MIMO demodulation*Final Project in Computer-aided VLSI System Design*

10/2023 - 12/2023

Taipei, Taiwan

- Implemented QR decomposition engine with 2.2% soft LLR error rate in Verilog.
- Achieved 206 MHz operating frequency with 37.5 mW power under TSMC 130nm technology with full VLSI design flow (RTL to GDS).

AWARDS AND HONORS

ICCAD Student Scholar Program Grant, ICCAD

10/2025

Taiwan Chip-based Industrial Innovation Program Research Scholarship, NSTC

09/2024 - 01/2025

Scholarship for Overseas Internship Program, NTU / Ministry of Education

06/2024 - 08/2024

Research Scholarship, MediaTek-NTU Research Center

03/2024 - 06/2024

Second Prize and Global Nominee, NASA Space App Challenge

10/2023

College Student Research Scholarship, NSTC

08/2023 - 02/2024

PROFESSIONAL SERVICE

Reviewer for 45th International Conference on Computer-Aided Design (ICCAD'26)

Reviewer for 34th International Symposium on Physical Design (ISPD'25)

Volunteer for 27th International Conference on Quantum Information Processing (QIP'24)

Volunteer for the First IEEE International Workshop on LLM-Aided Design (LAD'24)